

SS1 Science Track: Biology Diagnostic (Part 1)

Standard: Cambridge IGCSE / O-Level Bridge

Cluster 1: Cell Biology, Transport & Human Systems (Questions 1 - 20)

No.	Question & Options	Ans	Diagnostic Focus
1	Which organelle is responsible for cellular respiration and energy production? A) Nucleus B) Ribosome C) Mitochondrion D) Chloroplast	C	Cell Organelles & Function.
2	Which structure is found in a plant cell but not in an animal cell? A) Cell membrane B) Cytoplasm C) Cell wall D) Nucleus	C	Cell Differentiation.
3	The movement of water molecules from a region of high concentration to low concentration across a semi-permeable membrane is: A) Diffusion B) Osmosis C) Active Transport D) Transpiration	B	Transport Mechanisms (Osmosis).
4	Which process requires energy (ATP) to move substances against a concentration gradient? A) Osmosis B) Diffusion C) Active Transport D) Facilitated Diffusion	C	Active vs. Passive Transport.
5	Enzymes are biological catalysts made of: A) Carbohydrates B) Lipids C) Proteins D) Nucleic Acids	C	Biochemistry (Macromolecules).
6	What happens to an enzyme when it is exposed to very high temperatures? A) It works faster B) It becomes denatured C) It becomes more active D) It freezes	B	Enzyme Kinetics (Denaturation).
7	The process by which green plants make their own food using sunlight is: A) Respiration B) Photosynthesis C) Transpiration D) Fermentation	B	Plant Nutrition.
8	Which gas is a by-product of photosynthesis? A) Carbon Dioxide B) Nitrogen C) Oxygen D) Hydrogen	C	Photosynthesis Equation.
9	The human heart has how many chambers? A) 2 B) 3 C) 4 D) 5	C	Human Anatomy (Circulatory).
10	Which blood vessel carries oxygenated blood away from the heart? A) Vein B) Artery C) Capillary D) Vena Cava	B	Circulatory System Logic.
11	The functional unit of the kidney is the: A) Neuron B) Nephron C) Alveolus D) Villus	B	Excretory System.
12	Which part of the brain controls balance and coordination? A) Cerebrum B) Cerebellum C) Medulla Oblongata D) Hypothalamus	B	Nervous System.
13	The exchange of gases in the lungs takes place in the: A) Trachea B) Bronchi C) Alveoli D) Diaphragm	C	Respiratory System.

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14	Which of these is a characteristic of living things? A) Movement B) Respiration C) Sensitivity D) All of the above	D	MRS GREN (Characteristics of Life).
15	The basic unit of classification is the: A) Kingdom B) Phylum C) Genus D) Species	D	Taxonomy.
16	Which of these is a prokaryote? A) Fungi B) Bacteria C) Plants D) Animals	B	Cell Types (Prokaryote vs Eukaryote).
17	The process of breaking down food into absorbable molecules is called: A) Ingestion B) Digestion C) Egestion D) Absorption	B	Digestive System.
18	Which vitamin is essential for blood clotting? A) Vitamin A B) Vitamin C C) Vitamin D D) Vitamin K	D	Nutrition & Health.
19	The 'lock and key' hypothesis is used to explain: A) Osmosis B) Enzyme action C) Photosynthesis D) Respiration	B	Biological Models.
20	Diagnostic Insight: Students who cannot differentiate between Osmosis and Active Transport (Q3 vs Q4) will struggle with the physiological demands of SS1 Biology.	-	Conceptual Clarity Check.